Revealing the new arboreal frog genus *Auparoparo* after 130 years of *Oreophryne* sensu lato, subfamily Asterophrinae

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Abstract

Oreophryne sensu lato (Boettger 1895) is the largest clade of the hyperdiverse microhylid frog subfamily Asterophryinae, and is of particular concern for revision. *Oreophryne* is a large genus (73 or 75? species) with a surprisingly expansive distribution across Papua New Guinea to Southeast Asia, representing nearly all of the arboreal species in the microhylid subfamily Asterophryinae. A recent molecular phylogeny has demonstrated that Oreophryne as currently recognized is polyphyletic and consists of two reciprocally monophyletic genera, one of which is highly supported as being the most basal of the entire subfamily whereas the other unrelated genus is ~2M years younger. Our acquisition of four samples from North Maluku Island, Indonesia, including a specimen of the type species, *O. senckenbergiana*, has now allowed for a decisive resolution as to which clade can bear the name *Oreophryne*. We sequenced five loci (three nuclear: BDNF, SIA, NXC, and two mitochondrial: CytB, ND4), aligned them to the 236 tip Asterophryinae molecular dataset of Hill et al. (2022) and regenerated the phylogeny using Maximum Likelihood inference in IQTREE2. The type species fell into the more basal clade ("Oreophryne A" in Hill et al. 2022) indicating that the true *Oreophryne* is the sister branch to the rest of Asterophryinae. All four North Maluku island specimens were closely related and formed a monophyletic clade, which diverged from sister taxa on New Guinea Island about 10 MYA. We propose the name *Auparoparo* for the new genus (the younger clade Oreophryne B in Hill et al. 2022). These genera remain morphologically similar and we provide an analysis of their morphological traits and distribution in addition to redefinition. Our findings clearly demonstrate that arboreality has evolved at least twice within Asterophryne, making these two clades strong candidates for convergent evolution.

More complete sampling of additional species with greater range can help to further inform the dynamics of diversification of these ecologically diverse clades.

Introduction

The frog genus *Oreophryne* sensu lato (Boettger 1895) is the largest genus of the hyperdiverse Microhylidae:Asterophryinae, the largest subfamily of Microhylidae. With 75 diagnosed species, *Oreophryne* sensu lato makes up one fifth of the currently ~365 recognized species of Asterophryinae (Hill et al. 2022; Frost 2024), despite being only one of ~16 genera. *Oreophryne* sensu lato are conspicuously arboreal (or scansorial) and the genus accounts for nearly all arboreal species of Asterophryinae (Günther et al. 2001; Menzies 2006). *Oreophryne* sensu lato also has the largest geographic distribution of any genus in its subfamily, and is therefore an important group for understanding the biogeography of the region (Hill et al. 2023a). *Oreophryne* sensu lato occupy mostly low to mid-elevation ranges with species centered on mainland New Guinea (Zweifel et al. 2003; Frost 2024) and its offshore islands including the D Entrecasteaux and Louisiade Archipelagos (Zweifel 1956; Kraus and Allison 2004; Kraus and Allison 2009; Kraus 2013), New Britain Island (Werner 1898; Zweifel et al. 2003), but ranging westward to Sulawesi (Müller 1894; Boulenger 1896; Ahl 1933; Putri et al. 2023), northward to Indonesia (Boettger 1895; Setiadi et al. 2010; Günther et al. 2023) and the Southern Philippines Islands (Alcala 1986; Balinton and Seña 2015). Several species are also known at high elevation where they are no longer arboreal, and instead adopt a terrestrial lifestyle (Zweifel et al. 2009; Günther and Richards 2011; Putri et al. 2023).

Although the generic taxonomy of the subfamily Asterophryinae has had a turbulent history (reviewed in Hill et al. 2022), the taxonomy of the genus *Oreophryne* sensu lato in particular has been comparatively stable until recently. Generic designation to *Oreophryne* sensu lato has traditionally relied on a pectoral anatomy that is distinct from all other genera, with highly reduced and distinctly curved clavicles (Parker 1934; Zweifel 2003). However, this trait has never been demonstrated to be a synapomorphy. Hill et. al (2022, 2023b) established that *Oreophryne* sensu lato represents two reciprocally monophyletic groups, temporarily labeled "Oreophryne A" and "Oreophryne B". Because the type species *Oreophryne senckenbergiana* has not yet been included in large genus-level molecular phylogenies, it has not been possible to decipher which of the two groups can retain the genus name *Oreophryne*.

Taxonomic History:

Boettger (1895) described the genus *Oreoprhyne* sensu lato and the species *Oreophryne senckenbergiana* on the basis of two specimens from Halmahera Island, in the Maluku Archipelago west of New Guinea, that lacked procoracoids and had cartilaginous sterna. He also noted the presence of expanded finger pads and transverse folds of skin in the palate. Parker (1934) noted that procoracoids were present, but were reduced and re-diagnosed the genus on the basis of this feature together with the presence of reduced clavicles, no omosternum and a large cartilaginous sternum. Zweifel (2003) noted that "the clavicles are tiny, slightly curved bones lying on (ventral to) the procoracoid cartilage and apart from the midline and the scapula." On the basis of these features there are now 75 recognized species in the genus (Frost 2024). Most of these are small, arboreal or scansorial frogs that inhabit lowland to montane rain forest, but there is a small group of diminutive species, some of which are terrestrial, that inhabit high elevation grasslands (Zweifel et al. 2009).

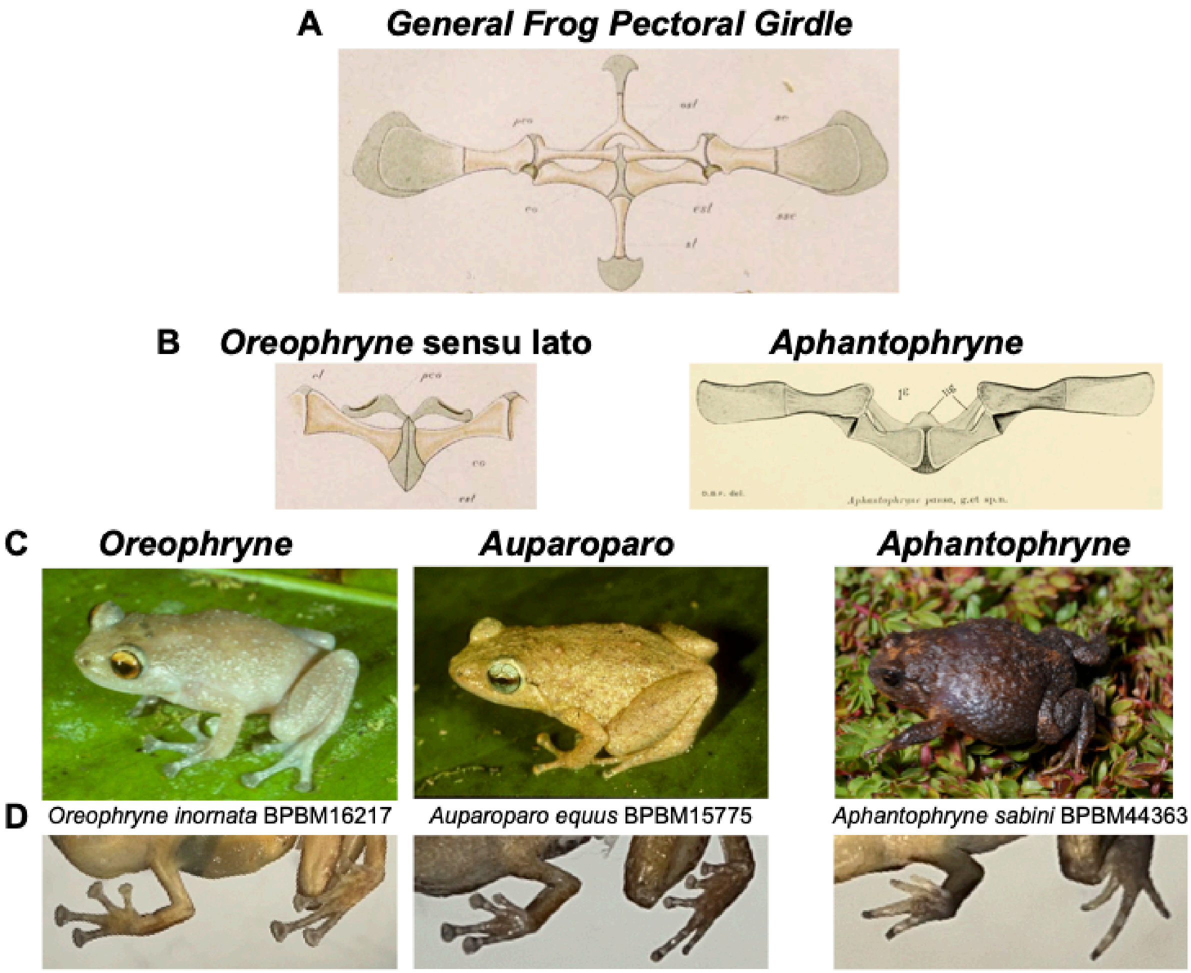

Figure 1: The morphology of *Oreophryne* and *Auparoparo*. (A) the generalized pectoral girdle of anurans as exemplified by *Cornufer corrugatus*, from Méhely (1897), (B) the pectoral girdle *Oreophryne biroi*, as an example of *Oreophryne* and *Auparoparo*, (C) photos of *Oreophryne inornata* and *Auparoparo equus* in life, and (D) toe pads of the fore and hind limbs.

While *Oreophryne* sensu lato is united morphologically, large scale molecular phylogenetic analysis with many ingroup samples were required to clarify that this group contains two reciprocally monophyletic clades (Hill et al. 2022, 2023b). Early molecular phylogenetic studies suggested monophyly but only included a few taxa within a larger-scale study (Köhler and Günther 2008 included 9 Oreophryne taxa; de Sa et al. 2012 included 4). Rivera et al. (2017) provided the first molecular evidence of a polyphyletic *Oreophryne* sensu lato with 36 *Oreophryne* within a larger phylogeny Asterophryinae which included 155 samples, followed by Tu et al. (2018)'s sparse supermatrix approach with 18 taxa of Oreophryne falling out in various places within a much larger phylogeny, but both of these lacked strong nodal support connecting the hypothesized clades to the backbone of the phylogeny. The polyphyletic nature of *Oreophryne* sensu lato was not clearly demonstrated until Hill et al. (2022),(2023b) included 44 taxa of *Oreophryne* with complete data sampling within a 218 taxon phylogeny of Asterophryinae. This phylogeny is the most complete and robust phylogeny to date. The two clades, labeled "Oreophryne A" and "Oreophryne B" have deep histories of ~ 18MY and ~16MY, respectively, with "Oreophryne A" most basal to the rest of the subfamily.

The aims of this study are to include the type species of *Oreophryne* (*Oreophryne senkenbergiana*) and its close sister taxon (*Oreophryne frontifasciata*) into a large Asterophrinae (subfamily-level) phylogeny, to develop a phylogenetic definition of *Oreophryne* and the new genus, and to name and describe the new genus. Furthermore, we analyze the morphological characteristics and distribution of these genera to assist collectors when identifying putative *Oreophryne* in the field, and discuss the evolutionary patterns and taxonomic utility of these morphological characters.

Materials and Methods

NOTE: M&M IS FROM THE NEW OREOPHRYNE MANUSCRIPT - CHECK IF ANYTHING NEEDS TO BE BROUGHT OVER FROM THE GOOGLE DOC.

Taxon Sampling

Asterophryinae is an adaptive radiation with short branch lengths along the backbone separating generic-level taxa (Rivera et al. 2017; Hill et al. 2022, 2023b), we reconstruct the phylogeny of *Oreophryne* sensu lato within the larger context of the subfamily. We include 30 species and 18 unnamed taxa of *Oreophryne* sensu lato within a larger dataset of 235 samples of 204 taxa of Asterophryinae collected across New Guinea, the Philippines, and Indonesia with three outgroup species to root the phylogeny (*Dyscophus antongilii*, *Scaphiophryne marmorata*, and *Platypelis grandis*). Importantly, we add the type species *Oreophryne senckenbergiana* (= *O. moluccensis*, Peters and Doria 1878) and the closely related *Oreophryne frontifasciata*, in a molecular phylogeny for the first time. Both taxa were collected in Halmahera, North Maluku Province, Indonesia by Iqbal Setiadi , Table 1, near the type locality for Oreophryne (Halmahera, Ternate, and Batjan Islands, Indonesia as documented by Peters and Doria 1878; Boettger 1895).

Table 1: Locality information for samples sequenced in this study. See Hill, et al., 2023
1 for additional metadata.

ID	Species	Locality
BJE01248	*Oreophryne frontifasciata*	Kabupaten Halmahera Timur, Halmahera Island, North Maluku
BJE01143	*Oreophryne senckenbergiana*	Kabupaten Halmahera Utara, Halmahera Island, North Maluku
BJE01144	*Oreophryne senckenbergiana*	Kabupaten Halmahera Utara. Halmahera Island, North Maluku
BJE01120	*Oreophryne senckenbergiana*	Kabupaten Halmahera Utara, Halmahera Island, North Ma

Molecular Sequence

We obtained nearly complete DNA sequence data for five loci (three nuclear: Seventh in Absentia [SIA], Brain Derived Neurotrophic Factor [BDNF], Sodium Calcium Exchange subunit-1 [NXC-1], and two mitochondrial loci (Cytochrome oxidase b [CYTB], and NADH dehydrogenase subunit 4 [ND4] from four samples of two taxa not previously sequenced and combined these with previously published data (Rivera et al. 2017; Hill et al. 2022, 2023b) for an alignment of 240 samples with 207 taxa and 2475 base pairs that is 96% complete. All loci were complete for the added samples except for CYTB from *Oreophryne frontifasciata*. Sequence quality was evaluated using chromatogram quality scores, checks for stop codons indicating pseudogenes, BLAST search (Ye et al. 2012), and concordance of pairwise similarity with other Asterophryinae sequences. Primer sequences and detailed methods are provided in (Hill et al. 2023b). We note that we excluded from the previous alignment (Hill et al. 2022, 2023b) two samples of *A. pansa* (BPBM 5299 and BPBM 8312) which were deemed unreliable as they came from formalin-fixed material, and three samples that should be confirmed with new material (BPBM 22708 *Copiula tyleri*, BPBM 38939 *Copiula sp. 7*, BPBM 37753 *Paedophryne dekot*). We note that genetic material from 46 species of *Oreophryne* sensu lato was not available for this molecular diagnosis.

Morphology, Behavior, and Geographic Distribution

We collected new morphological characters and reexamined traits compiled from the diagnosing literature , character states defined in 2 for all named species of *Oreophryne* in our phylogenetic dataset. Our sample included all species available in the collections of the Bishop Museum, Honolulu, Hawaii (BPBM): thirteen species of "Oreophryne A", eight species "Oreophryne B"

(Table 3). Finger and toe pads were photographed and length and width of the toe tips were measured using ImageJ software. Trait information for the remaining species were gathered from the literature, six species of "Oreophryne A" and one "Oreophryne B". In addition, we compiled information on call type from the literature and collection location from the respective collectors or the BPBM database (global positioning coordinates [GPS]; Table 3). We examined the ability of morphological and behavioral characters to provide a diagnosis to the genus-level clade consistent with molecular phylogenetics.

Table 2: Description of traits surveyed. Several traits are traditionally noted for Aste
nae species: presence of two palatal folds, horizontal pupil, and lack of vomeri
(Boettger 1895; Kampen 1923; Parker 1934).

Trait	Definition
Procoracoids	Size and shape of the procoracoid cartilages, if present
Clavicles	Size and shape of the clavicles, if present
Omasternum	Presence or absence of the omasternum
Sternum	Sternal plate, or cartilaginous sternum, if present
Call type	Qualitative description of call type
Finger:toe width	The ratio of finger pad width to toe pad width
Pectoral ligament	Type of connection between scapula and procoracoid, either ligamentous or ca
Rel toe length	Relative length of 5th toe compared to length of 3rd toe (5<3, 5=3, 5>3)
Toe webbing	Degree of webbing based on how high toe webs reach on the 5th toe (none, ba
Tympanum:eye diameter	Tympanum diameter relative to eye diameter (¼, □ , or ½)
Reference	Citation
Pupil horizontal	Whether the pupil is horizontal
Teeth	Whether teeth are present
Palatal folds	Whether two horizontal palatal folds in front of the pharynx are present

Nomenclature

We conducted a comprehensive taxonomic literature review of all diagnoses for species known to belong to the clades "Oreophryne A" and "Oreophryne B" (Hill et al. 2022). We applied the ICZN code (Zoological Nomenclature et al. 1999) as well as the Phylocode (de Queiroz and Cantino 2020). We support the nascent effort to assign indigenous names in taxonomy whenever possible (e.g., Gillman and Wright 2020). One of us (B. Iova) is a local indigenous language expert, and provided guidance in proposing the novel genus name, *Auparoparo*, meaning "tree frog" in Motu, a regional language, for *Oreophryne* "B".

Phylogenetic Reconstruction

We used Partitionfinder in IQTREE2 to find the best-fit evolutionary model followed by Maximum Likelihood (ML) phylogenetic reconstruction (Lanfear et al. 2012; Minh et al. 2020). We tested models partitioned by both locus and codon versus by loci only (15 vs. 5 partition scheme), and found that the locus and codon model was superior by the Akaike Information Criterion. Nodal support values for the ML tree were inferred using 2000 bootstrap replicates.

Phylogenetic Definition

We use a maximum crown-clade definition under the PhyloCode (de Queiroz and Gauthier 1990, 1992; de Queiroz and Cantino 2020), specifically, the largest crown clade originating with the most recent common ancestor of the reference species for the clade (e.g., the type species of the genus), and all extant species of the clade more recently than they do with reference species of other clades (i.e., in this study, other generic-level clades). This definition is robust to uncertainty in basal relationships such as in the case of adaptive radiations with short branch lengths along the backbone as in Asterophryinae, as well as being robust to identification of new species and new clades.

Analyses, Visualization, and Data Availability

All analyses were conducted and graphics produced in the R Statistical Computing Environment unless noted below (R Core Team 2024). Initial DNA sequence alignment was conducted using `MUSCLE` (Edgar 2004) with manual adjustments in `Mesquite` (Maddison and Maddison 2023). Phylogenetic reconstruction was conducted using the `IQTREE2` suite of software (Lanfear et al. 2012; Minh et al. 2020).

The correspondence of the phylogenies to geographic map locations with the `phylo.to.map` function, both from the `phytools` package (Revell 2012). All other tree graphics were produced with the `ggtree` package Yu (2020) supplemented by `ggplot2` (Wickham 2016). All data, alignments, and code to reproduce analyses are available in the public repository ...

Results

Phylogeny and Phylogenetic Taxonomy

A partition scheme including codon as well as locus resulted in an improvment of fit by 3,127 AIC units (evolutionary model partitioned by locus and codon, 15-partitions: AIC of 155,492, versus partitioned by locus only, 5 partitions: AIC of 158,619). Unless otherwise noted, all phylogenetic reconstructions used the 15-partition scheme identified as best-fit by IQTREE2.

We recovered a maximum likelihood topology that confirms that *Oreophryne* sensu lato is polyphyletic, with species falling into two independently-evolved, reciprocally monophyletic, highly-supported clades Figure 2. Nearly all nodes are moderately or highly supported, with most of the weakly supported nodes falling along the backbone of the adaptive radiation with short branch lengths.

Furthermore, these findings demonstrate that the clade descended from node A should retain the converted clade name (CCN) *Oreophryne*, as the type species, *O. senckenbergiana*, as well as a closely-related sister species *O. frontifasciata*, both from North Maluku Island, Indonesia, are situated within *Oreophryne* with high support (Figure 2). We propose the new clade name (NCN) *Auparoparo* for the node descended from node B. *Oreophryne* is the oldest clade in the Papuan subfamily, with genera *Cophixalus* and *Paedophryne* arising before *Auparoparo*, which split from the remaining genera of Asterophryinae.

Morphology, Behavior, and Ranked Taxonomy

We find no morphological synapomorphies that distinguish *Auparoparo* from *Oreophryne*. *Oreophryne* sensu lato is diagnosed by characters of the pectoral girdle: posessing highly reduced, curved procoracoid cartilages and clavicles, and lacking an omosternum (Parker 1934; Zweifel 2003). We found that this suite of characters are present in both *Oreophryne* and *Auparoparo*, and therefore is not diagnostic for neither *Oreophryne* nor *Auparoparo* (Table 3).

Two traits imperfectly segretate between the clades, toe pad widths and call type (Table 3), which may assist in preliminary identification. *Oreophryne* is an arboreal clade with expanded digital pads of roughly equal width between the fingers and toes, whereas *Auparoparo* have visibly larger finger pads, with finger pads roughly 20-30% wider than toes. Both *Oreophryne* and *Auparoparo* use a variety of call types, with “honks” or “peeps” used only by *Oreophryne*, whereas “whinny” calls are used only by arboreal *Auparoparo*. A minority of species in both groups use the “rattle” call type. Both toe pad width and call type are variable traits with some overlap and thus not reliable for diagnosis. Furthermore, toe pad is clearly an adaptive trait for arboreality.

Genus	Species	Procoracoids	Clavicles	Omosternum	Sternum
	anamiatoi	curved reduced	curved reduced	absent	cartilagino
	anulata[1]	curved reduced	curved reduced	absent	cartilagino
	aurora	curved reduced	curved reduced	absent	cartilagino
	biroi	curved reduced	curved reduced	absent	cartilagino
	brachypus	curved reduced	curved reduced	absent	cartilagino
	cameroni	curved reduced	curved reduced	absent	cartilagino
	frontifasciata[1]	curved reduced	curved reduced	absent	cartilagino
	geislerorum	curved reduced	curved reduced	absent	cartilagino
	graminis[1]	curved reduced	curved reduced	absent	cartilagino
	inornata	curved reduced	curved reduced	absent	cartilagino
	lemur	curved reduced	curved reduced	absent	cartilagino
	loriae	curved reduced	curved reduced	absent	cartilagino
	nana[1]	curved reduced	curved reduced	absent	cartilagino

Genus	Species	Procoracoids	Clavicles	Omosternum	Sternum
	notata	curved reduced	curved reduced	absent	cartilagino
	parkeri	curved reduced	curved reduced	absent	cartilagino
	parkopanorum	curved reduced	curved reduced	absent	cartilagino
	philosylleptoris	curved reduced	curved reduced	absent	cartilagino
	senckenbergiana[1]	curved reduced	curved reduced	absent	cartilagino
Auparoparo[2]	*brunnea*[1]	curved reduced	curved reduced	absent	cartilagino
	equus	curved reduced	curved reduced	absent	cartilagino
	ezra	curved reduced	curved reduced	absent	cartilagino
	insulana	curved reduced	curved reduced	absent	cartilagino
	matawan	curved reduced	curved reduced	absent	cartilagino
	meliades	curved reduced	curved reduced	absent	cartilagino
	penelopeia	curved reduced	curved reduced	absent	cartilagino
	phoebe	curved reduced	curved reduced	absent	cartilagino
	picticrus	curved reduced	curved reduced	absent	cartilagino
	variabilis[1]	curved reduced	curved reduced	absent	cartilagino

[1]Character states obtained from the literature, specimens not available for examination in this study.

[2]Hill et al. 2022 places *Aphantophryne pansa* within *Auparoparo*, but the status of *Aphantophryne* requires fu

*External characters were examined from specimens, internal characters and pupil shape taken from the liter

Geographic Distribution

The geographic distributions of *Auparoparo* and *Oreophryne* are shown in Figure 3, with corresponding phylogenetic relationships.

Taxonomy

We use a maximum crown-clade definition for designating *Oreophryne* and *Auparoparo* because the monophyly of these taxa are strongly supported, and the relationships of these taxa to each other and to sister genera of Asterophryinae are well-supported (this study, Hill et al. 2022,

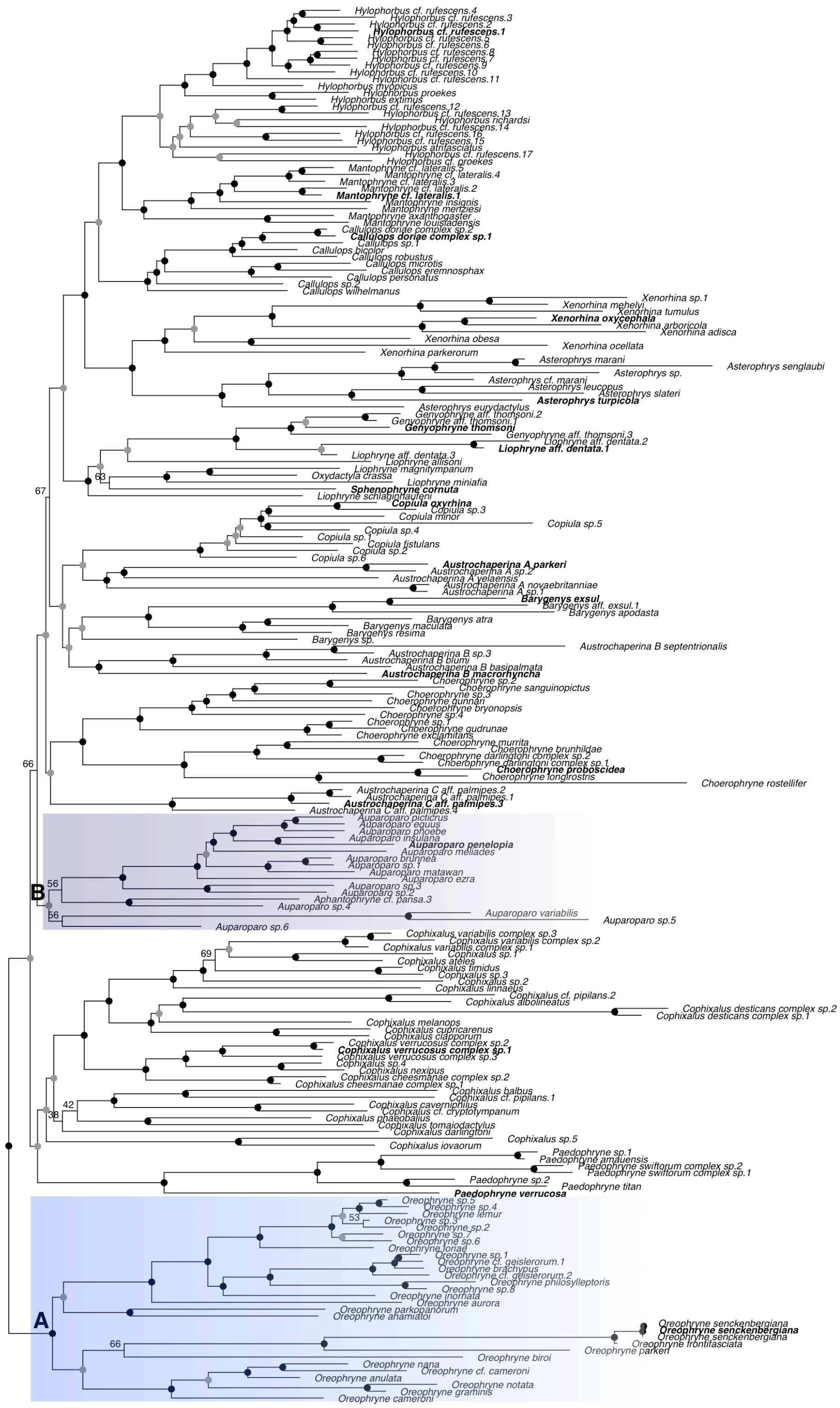

Figure 2: Molecular phyogenetic hypothesis of Asterophrinae showing polyphyly of *Oreophryne* sensu lato, into two reciprocally monophyletic clades, node A is *Oreophryne* and node B is *Auparoparo* based on the position of the type species *Oreophryne senkenbergiana*. Based on two mitochondrial and three nuclear markers following Rivera et al. (2017); Hill et al. (2022). Nodal support: black dots indicate ML boostrap support >= 95%, grey dots >= 75%, numeric values given for nodes below 75% support. Outgroup taxa excluded.

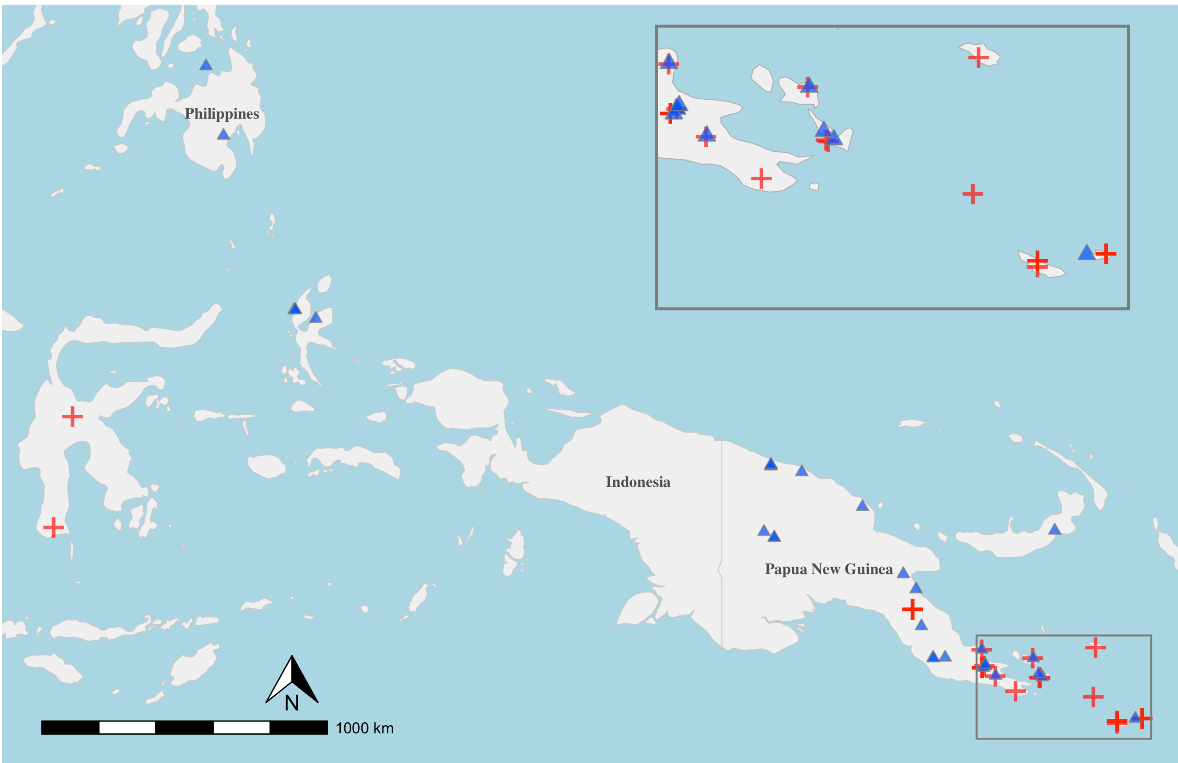

Figure 3: Distributions of and *Auparoparo* (red) and *Oreophryne* (blue).

2023b), as has been done for other anuran taxa (Brown et al. 2015) and summarized in the PhyloCode (de Queiroz and Gauthier 1990, 1992, 1994). That is, we adopt a nodal definition of the clade that identifies the genus as a monophyletic clade. Specifically, to include the type species of the genus and the most recent common ancestor that includes all extant species of the genus but not members of other genera. This definition is robust to uncertainty in basal relationships such as in the case of adaptive radiations with short branch lengths along the backbone as in Asterophryinae, as well as being robust to identification of new species. We follow Brown et al. (2015) in using the terms converted clade name (CCN) to indicate the phylogenetic redefinition of Oreophryne, and new clade name (NCN) for newly designated names (*Auparoparo*), whereas the previously defined polyphyletic *Oreophryne* is indicated by "sensu lato".

Family Microhylidae Günther, 1858 (1843)
Subfamily Asterophryinae Günther, 1858
Genus Oreophryne Boettger, 1895 amended

Microhyla Peters and Doria, 1878
Callula
Oreophryne Boettger, 1895: 135. Type species: *Oreophryne senckenbergiana*
Boettger, 1895 (= Microhyla achatina var. moluccensis Peters and Doria, 1878), by monotypy. Placed on Official List of Generic Names in Zoology by Opinion 1266, Anonymous, 1984.
Sphenophryne
Phrynixalus
Mehelyia Wandolleck, 1911 "1910", Type species: *Meheylia lineata* Wandolleck, 1911 (= *Sphenophryne biroi* Méhely, 1897), by subsequent designation of Parker, 1934. Synonymy by Nieden, 1926.
Cophixalus
Chaperina

Type species Oreophryne senckenbergiana (Peters and Doria, 1878)

Diagnosis: We note that there are no synapomorphies in adult *Oreophryne* that differ from the morphologically similar genus *Auparoparo*. Diagnoses should be made by molecular phylogenetic analysis, based on the phylogenetic definition below.

Description: *Oreophryne* differs from most other genera in subfamily Asterophryinae, but not from *Auparoparo*, by the combination of (1) presence of highly reduced clavicles (vs absence or elongate) which have a distinctly curved shape (Figure 1); (2) presence of well developed digital pads, often with finger and toe pads of similar size; (3) arboreal lifestyle (perch height >2m above ground); (4) most species range in body size from 16mm (*O. graminis*, Günther and Richards 2011) to 30mm, with a few larger species ~40mm (*O. inornata*); (5) small to invisible tympanum; (6) large eyes with pupil horizontal; (7) occasional presence of toe webbing; (8) thin, elongated limbs; and (9) in many species males that call with "honk" or "peep" types or in some species "rattle" type. Some species of *Oreophryne* may be distinguished from *Auparoparo* by roughly equal finger to toe pad ratio and males calling with "honk" or "peep" but not "whinny" call types.

Phylogenetic definition: *Oreophryne* (CCN) is defined as the maximum crown-clade (the monophyletic clade descending from node A, Figure 2) originating with the most recent common ancestor of *Oreophryne senckenbergiana* (Peters and Doria 1878), the type species of *Oreophryne*, and all extant species that share a common ancestor with *O. senckenbergiana* more recently than they do with reference species of all the other nominal generic clades listed in Table 4; with reference species of the closest sister clades to *Oreophryne* currently being *Auparoparo penelopeia* Kraus (2016), *Cophixalus verrucosus* Boettger (1892), and *Paedophryne kathismaphlox* (**Kraus:2010?**).

Table 4: Genera of Asterophryinae and their type species.

Genus	Type Species
Asterophrys	Asterophrys turpicola Schlegel 1837
Auparoparo	Auparoparo penelopeia Kraus 2016
*Austrochaperina**	Austrochaperina palmipes Zweifel 1956
*Austrochaperina**	Austrochaperina robusta Fry 1912
Barygenys	Barygenys cheesmanae Parker 1936
Callulops	Callulops doriae Boulenger 1888
Choerophryne	Choerophryne proboscidea Van Kampen 1914
Cophixalus	Cophixalus verrucosus Boettger 1892
Copiula	Copiula oxyrhina Boulenger 1898
*Genyophryne**	Genyophryne thomsoni Boulenger, 1890
Hylophorbus	Hylophorbus rufescens Macleay 1878
Mantophyrne	Mantophryne lateralis Boulenger 1897
*Liophryne**	Liophryne rhododactyla Boulenger 1897
Oreophryne	Oreophryne senckenbergiana (Peters and Doria, 1878)
Oxydactyla	Oxydactyla brevicrus Van Kampen 1913
Paedophryne	Paedophryne kathismaphlox Kraus 2010
*Sphenophryne**	Sphenophryne cornuta Peters and Doria 1878
Xenorhina	Xenorhina oxycephala Schlegel 1858

Genus	Type Species

*We note that the monophyly of Austrochaperina, Liophryne, Oxydactyla, and Sphenophryne (and reciprocal monophyly of Genyophryne [Genyophryne thomsoni Boulenger, 1890] with respect to Liophryne+Oxytactyla+Sphenophryne) has not yet been established.

Content: Our results support the retention of 18 species in *Oreophryne* with molecular evidence: *Oreophryne anamiatoi* (Kraus & Allison, 2009), *O. anulata* ((Stejneger 1908)), *O. aurora* (Kraus, 2016), *O. biroi* (Méhely, 1897), *O. brachypus* (Werner, 1898), *Oreophryne cameroni* ([Kraus, 2013]), *O. frontifasciata* ((Horst, 1883), *O. geislerorum* (Boettger, 1892), *O. graminis* (Günther & Richards, 2012), *Oreophryne inornata* (Zweifel 1956), *Oreophryne lemur* (Kraus, 2016), *O. loriae* (Boulenger, 1898), *O. nana* (Brown and Alcala, 1967), *O. notata* (Zweifel, 2003), *Oreophryne parkeri* (Loveridge, 1955), *Oreophryne parkopanorum* (Kraus, 2013), *Oreophryne philosylleptoris* (Kraus, 2016), and *O. senckenbergiana* (Peters & Doria, 1878). A number of species were not available for our study but should molecular evidence demonstrate their relationship within Oreophryne, those species should also retain their generic identity.

Distribution: The range of confirmed species in this dataset remains nearly as broad as previously defined (Figure 3), from the Philippines and Indonesia to the southeastern most tip of mainland New Guinea and several near-shore islands (Fergusson, and Normanby Is.) and the distant Rossel Island of the Louisiade Archipelago. However, the transfer of species which occupy Woodlark Islands and Misima and Sudest Islands of the Louisaide Archipelago reduces the known range of Oreophryne. Woodlark Island is inhabited by *O. phoebe* [=*Auparoparo phoebe*], Misima Island is inhabited by *O. picticrus* [=*Auparoparo picticrus*], and Sudest Island is inhabited by *O. ezra* and *O. meliades* [= *Auparoparo ezra* and *Auparoparo meliades*].

Family Microhylidae Günther, 1858
Subfamily Asterophryinae Günther, 1858
Genus Auparoparo gen. Nov.
Oreophryne Boettger, 1895
Aphantophryne Fry, 1917
Sphenophryne (anthonyi) Boulenger, 1897
Type species *Oreophryne penelopeia* (Kraus 2016) [=*Auparoparo penelopeia*]

Diagnosis: We note that there are no synapomorphies in adult *Auparoparo* that differ from the morphologically similar genus *Oreophryne*. Diagnoses should be made by molecular phylogenetic analysis, based on the phylogenetic definition below.

Description: *Auparoparo* differs from most other genera in subfamily Asterophryinae, but not from *Oreophryne*, by the combination of (1) presence of highly reduced clavicles (vs absence or elongate) which have a distinctly curved shape; (2) presence of well developed digital discs, with finger discs notably wider than toe discs (typically 20% or more); (3) arboreal lifestyle (perch height >2m above ground); (4) small to moderately large body size in known species, from 16mm (*A. brunnea*, Kraus 2017) to 25 mm (A. picticrus, Kraus 2016); (5) small to invisible tympanum; (6) large eyes with pupil horizontal; (7) occasional presence of toe webbing; (8) and thin, elongated limbs; and (9) and in many species males that call with a "whinny" call type or in some species "rattle" type. Some species of *Auparoparo* may be distinguished from *Oreophryne* by larger finger pads than toe pads and males calling with "whinny" but not "honk" or "peep" call types.

Phylogenetic definition: *Auparoparo* (NCN) applies to the largest crown clade originating in (CCN) is defined as the maximum crown-clade originating with the most recent common ancestor of *Auparoparo penelopeia* (Kraus 2016), the type species of *Auparoparo*, and all extant species that share a common ancestor with *A. penelopeia* more recently than they do with the type species of all the other nominal genera listed in Table 4, the type species of the closest sister taxa being *Oreophryne senckenbergiana* (Peters and Doria 1878), *Cophixalus verrucosus* Boettger (1892), and *Paedophryne kathismaphlox* (**Kraus:2010?**).

Content: We transfer eleven named species from Oreophryne into *Auparoparo* gen. Nov.: *Auparoparo brunnea* (Kraus, 2017), *Auparoparo equus* (Kraus, 2016), *Auparoparo ezra* Kraus and Allison, 2009, *Auparoparo insulana* Zweifel, 1956, *Auparoparo matawan* Kraus, 2016, *Auparoparo meliades* Kraus, 2016, *Auparoparo penelopeia* Kraus, 2016, *Auparoparo phoebe* Kraus, 2017, *Auparoparo picticrus* Kraus, 2016, *Auparoparo nana*, *Auparoparo penelopia* (Kraus 2016), *Auparoparo variabilis* (Boulenger, 1896).

Distribution: Southeastern tip of mainland Papua New Guinea and satellite islands (Fergusson, Misima, Normanby, Rossel, Sudest, Woodlark Is.), Sulawesi.

Etymology: The generic name *Auparoparo* comes from the indigenous language, Motu, which is widely spoken in the region of Papua New Guinea in which this genus is found. It is a combination of the words "au" (tree) and "paro paro" (frog).

Discussion

Phenotypic Characters and Diagnoses

Our phylogenetic results (Hill et al. 2022; 2023b, this study) provide clarity on the classification of the problematic genus *Oreophryne* sensu lato. We confirm that *Oreophryne* sensu lato is polyphyletic and consists of two reciprocally monophyletic clades, "Oreophryne A" and "B" (of Hill et al. 2022), originating 18 and 16 MYA, respectively (Figure 1). Here we designate "Oreophryne A" as retaining the name *Oreophryne* based on its containing the type species *Oreophryne senckenbergiana* (Peters and Doria 1878) with high support. *Oreophryne* is the earliest diverged genus in subfamily Asterophryinae, sister to all the other genera. The second clade, "Oreophryne B", is highly supported as a separate, unrelated genus, which contains no known species with historical generic priority. Thus, we designate this genus as *Auparoparo*, a nominal tribute to the Motu language of the indigenous peoples of the region.

The Difficulty with Cryptic Genera

Perhaps the greatest testament to the crypticism of *Oreophryne* (CCN) vs. *Auparoparo* (NCN) is that no taxonomist was able to distinguish that they are separate clades, independently evolved, in the 130 years between their discovery and molecular sequencing (Boettger 1895; Rivera et al. 2017). These species are indeed morphologically very similar and our results provide some clues as to why *Oreophryne* taxonomy has been so difficult. We found no morphological synapomorphy for neither *Auparoparo* nor *Oreophryne*, and moreover, no morphological feature that can diagnose either clade or both in combination.

The traditional basis for generic designation of species to *Oreophryne* has primarily relied on a signature pectoral anatomy (whether original or revised diagnosis, Parker 1934; Zweifel 2003). Our phylogenetic results clearly show that this trait appears more than once, in both *Oreophryne* and *Auparaparo*, and rather than being a highly conserved trait, it is either a symplesiomorphy (shared ancestral character) or a result of convergent or parallel evolution, and perhaps adaptive for arboreality. Thus, the pectoral anatomy is not a reliable diagnostic character in differentiating between *Oreophryne* and *Auparoparo*. In studies across a wide diversity of anurans, Soliz et al. (2025) found that anatomical variation of the pectoral girdle corresponds with functional and ecological lifestyle demands, whereas Engelkes et al. (2020) found strong phylogenetic influence in the shape of the pectoral girdle. This point would be excellent for further study within Asterophryinae, as there are high elevation terrestrial species of *Oreophryne* sensu lato that possess the same pectoral trait (Zweifel et al. 2009),as well as a few arboreal species within other genera (*Cophixalus*, *Asterophrys*, *Xenorhina*) with unknown pectoral morphology.

We further rexamined additional characters previously used to diangnose *Oreophryne* as well as characters conventionally used to diagnose species of frogs. We conclude that pectoral girdle connector type (Parker 1934) and relative lengths of the 3rd and 5th toe are uninformative and the degree of toe webbing is subjective. It was previously suggested that species in *Oreophryne* sensu lato could be divided into two groups based on the type of connection between the procoracoid cartilages and the scapula (Kraus 2013), with roughly half possessing a cartilaginous rod extending to the scapula and others a ligamentous connection (Parker 1934). However, our phylogenetic evidence shows that these traits readily evolve and are not synapmorphies and therefore not reliable indicators of monophyletic groups. Ligamentous connectors are common across both genera (Table 3). Of species with toe webbing, *Oreophryne* tends to be described as having "well webbed" toes compared to *Auparoparo*'s "basal" webbing (Table 3). However, intra-specific variability combined with the subjective nature of these descriptions and inconsistency between specific markers when used ("reaching the penultimate tubercle", "webbed 1/3 length of 5th toe", etc.), makes this feature difficult to standardize for generic diagnostic use. Many of the morphological traits proposed by Parker (1934) and Zweifel

(2003) may in fact be a result of homoplasy to evolutionary pressures imposed by lifestyle or parallel evolution, and while they may prove useful for ecomorphological identification or species diagnoses in combination with other traits, are not informative for generic diagnosis.

We found only two characters with potential for imperfectly identifying genera: relative digital pad size (Parker 1934; Zweifel 1956), and call type (e.g., Kraus 2016). We found that the relative widths of the digit pads differs between genera: *Auparoparo* tends to have visibly larger finger pads in comparison to toe pads, whereas in *Oreophryne* the finger and toe pads are roughly equal in size, although there is some overlap in finger to toe pad ratio (Table 3). In some cases, *Auparoparo*'s finger pads display T-shapes while the toes are often rounder and narrower. Whether there is any performance advantage in larger finger pads compared to toes is unknown. Furthermore, a handful of high-elevation species not included in our phylogeny have evolved terrestrial or fossorial habits and possess smaller digital pads compared to their presumed arboreal relatives (Zweifel et al. 2009).

Call features are increasingly used for diagnosing species of *Oreophryne*, with advertisement call types commonly categorized as rattle, honk, whinny, peeping, or clicking (Kraus 2016). We found honk or peep call types are only found in *Oroephryne* species in our sample, while whinny calls are only found in *Auparoparo* species, and rattle calls are found in both genera (Table 3). We note that several authors described species of each genus, thus the difference between genera cannot be attributed to the investigator. However, more complete sampling may add further complexity. As far as we are aware there has been no comprehensive acoustic analysis across *Oreophryne* sensu lato. More research is needed to explore call diversity both between species and across genera to undertand call evolution and whether there any features that are distinctive of genera.

Geographic Distribution

Both *Oreophryne* and *Auparoparo* are geographically centered in the mainland New Guinea. *Oreophryne* extends southward to the Louisiade Archipelago and northward to the South Philippine Islands (Hill et al. 2023b). Addition of the type species *Oreophryne senkenbergiana* and *O. frontifasciata* expands the known range eastward to North Maluku, Indonesia (Figure 3).

The known distribution of *Auparoparo* is mainly clustered in the southeastern tip of mainland Papua New Guinea (in the East Papuan Composite Terrane) and satellite islands with two species in Sulawesi, Indonesia. This suggests the existence of yet undescribed or undiscovered species of *Auparoparo* with intermediate localities. This geographic range is sure to grow with additional phylogenetic sampling. For example *Oreophryne* (sensu lato) *chlorops* (Günther et al. 2023) is a very large (SVL $>$ 40mm) species recently discovered from mid-elevation mountains of the bird s neck region of Western Papua with clearly enlarged finger pads relative to toe pads. If confirmed as *Auparoparo*, it would significantly expand the range across New Guinea Island.

We found geographic range overlap between *Oreophryne* and *Auparoparo* at multiple localities: on the mainland along the Mt. Dayman Foothills, at the Mt. Simpson Massif, Mt. Trafalgar, and on the islands of Normanby and Fergusson (both nearshore islands), and Rossel island (hundreds of kilometers away), which presents interesting questions regarding their ecological relationships. For example, *Oreophryne inornata* and *Auparoparo insulana* have been collected at the same site (Goodenough Island) and even the same microhabitats (Zweifel 1956). How can species coexist in the same niche? And what is the temporal history of each colonization? Are both genera also present in the nearby islands of Normanby and Fergusson (all part of the D'Entrecasteaux Archipelago)? Obtaining additional geographical samples for inclusion in phylogenetic analysis should be a high priority to gain a more complete understanding of the geographic distribution and evolutionary history of both *Oreophryne* and *Auparoparo*.

Phylogenetic Taxonomy

We adopt an explicitly phylogenetic framework for the new taxonomy of the genera *Oreophryne* and *Auparoparo*, and to our knowledge, this is the first time that a phylogenetic taxonomic approach has been applied for any genus of the subfamily Asterophryinae. A nodal definition of genera is advantageous especially because Asterophryinae is an adaptive radiation. While we are increasingly confident that at least 16 genera are indeed monophyletic, nevertheless, very short branch lengths separate many of the genera along the backbone [Hill et al. (2022); this study]. A nodal definition is robust to uncertainty (i.e., rearrangements of order) along the backbone. We define genera as the largest crown clade originating with the most recent common ancestor of the reference species for the clade (e.g., the type species *Oreophryne senkenbergiana*), and all extant species of the clade more recently than they do with reference species of other clades that the sample may potentially belong to. The type species, if genetic material is available seems like a reasonable choice for the reference taxon, but for groups such as Asterophryinae, where a number of type species were collected in the 1800 s it may not be possible to obtain a genetic sample, as these locations may be uncertain and/or remote, and other proxies may be needed. Nevertheless, is increasingly clear that for *Oreophryne* and *Auparoparo*, a molecular phylogenetic approach is required for final confirmation of taxonomy.

Furthermore, a phylogenetic definition provides guidance for robust classification. Given the deep histories of the two genera, it is imperative that molecular confirmation of either *Oreophryne* or *Auparoparo* include members of the other clade as an outgroup, but also members of *Cophixalus*, *Paedophryne*, or other closest sister genera as outgroups to clearly resolve whether the sample shares an MRCA more closely with *Oreophryne* or *Auparoaro*. The ability to resolve the "boundaries" of the monophyletic groups requires it. An additional difficulty here is that within each genus, there are subclades with deep histories, so care must be taken to decisively demonstrate monophyly. For example, a phylogenetic analysis with only *Oreophryne* sensu lato and outgroups to Asterophryinae, will show grouping of *Oreophryne* and *Auparoparo*, because of lack of intervening phylogenetic history. Similarly, an analysis with only *Oreophryne* sensu lato with a member of one of the younger genera, say, *Hylophorbus*, will give a similar misleading result because both *Oreophryne* and *Auparoparo* are equally distant from *Hylophorbus*. Clearly a greater taxonomic representation is always better, but in this case, care is required to avoid a misleading result. This study furthermore highlights the importance of taxonomic sampling for robust phylogenetic results. Developing an orderly taxonomy in a clade with over 350 named species (and an estimated 40% more yet to be described species currently in museum collections) would be that much more challenging without the benefit of generic diagnoses in this interesting group.

Clarifying the generic structure also facilitates appreciation for the patterns of diversification: a tendency toward niche conservatism within genera, ecological and generic diversification occurring early in the history of the subfamily (Hill et al. 2022); and furthermore the very interesting biogeographical history of simultaneous dispersal of multiple genera to offshore islands (Hill et al. 2022, 2023a) revealing the very strong influence of plate tectonics for explaining the historical dispersal patterns of this group. Finally, without stability in the naming of genera and the independent histories embodied by each, it becomes much more difficult to appreciate that the two clades represented by *Oreophyne* and *Auparoparo* have independently evolved, that there are two independent arboreal clades, and would make it very difficult to appreciate the impressive convergence or paralleism embodied by these speciose and successful genera. An accurate and informative taxonomy serves to advance our understanding of this group and the very interesting evolutionary insights that they embody leading to further discovery.

Lastly, we emphasize that future naming of species, genera, and higher classifications should always seek to recognize the language of the indigenous stewards of the land and ecosystem from which they were discovered (Gillman and Wright 2020). This is especially true of Asterophryinae frogs, whose awe-inspiring diversity mirrors the rich cultural beauty and historical complexity of Papuan people.

Supplementary Table 1

Named species of Oreophryne sensu lato for which we do not have genetic material and cannot classify into Oreophryne vs. Auparoparo.

Oreophryne albitympanum (Günther, Richards, and Tjaturadi, 2018)
Oreophryne albomaculata (Günther, Richards, and Dahl, 2014)
Oreophryne albopunctata (Van Kampen, 1909)
Oreophryne alticola Zweifel, Cogger, and Richards, 2005
Oreophryne ampelos Kraus, 2011
Oreophryne anser Kraus, 2016
Oreophryne anthonyi (Boulenger, 1897)
Oreophryne asplenicola Günther, 2003
Oreophryne atrigularis Günther, Richards, and Iskandar, 2001

Oreophryne banshee Kraus, 2016
Oreophryne brevicrus Zweifel, 1956
Oreophryne brevirostris Zweifel, Cogger, and Richards, 2005
Oreophryne celebensis (Müller, 1894)
Oreophryne choerophrynoides Günther, 2015
Oreophryne chlorops Günther, Iskandar, and Richards, 2023
Oreophryne clamata Günther, 2003
Oreophryne crucifer (Van Kampen, 1913)
Oreophryne curator Günther, Richards, and Dahl, 2014
Oreophryne flava Parker, 1934
Oreophryne flavomaculata Günther and Richards, 2016
Oreophryne furu Günther, Richards, Tjaturadi, and Iskandar, 2009
Oreophryne geminus Zweifel, Cogger, and Richards, 2005
Oreophryne habbemensis Zweifel, Cogger, and Richards, 2005
Oreophryne hypsiops Zweifel, Menzies, and Price, 2003
Oreophryne idenburgensis Zweifel, 1956
Oreophryne jeffersoniana Dunn, 1928
Oreophryne kampeni Parker, 1934
Oreophryne kapisa Günther, 2003
Oreophryne mertoni (Roux, 1910)
Oreophryne minuta Richards and Iskandar, 2000
Oreophryne monticola (Boulenger, 1897)
Oreophryne nicolasi Richards and Günther, 2019
Oreophryne oviprotector Günther, Richards, Bickford, and Johnston, 2012
Oreophryne pseudasplenicola Günther, 2003
Oreophryne pseudunicolor Günther and Richards, 2016
Oreophryne roedeli Günther, 2015
Oreophryne rookmaakeri Mertens, 1927
Oreophryne riyantoi Putri, Trilaksono, Kurniati, Engilis, and Hamidy, 2023
Oreophryne sibilans Günther, 2003
Oreophryne streiffeleri Günther and Richards, 2012
Oreophryne terrestris Zweifel, Cogger, and Richards, 2005
Oreophryne unicolor Günther, 2003
Oreophryne waira Günther, 2003
Oreophryne wapoga Günther, Richards, and Iskandar, 2001
Oreophryne wolterstorffi (Werner, 1901)
Oreophryne zimmeri Ahl, 1933

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